1. $$\frac{(8+2\cdot 0-3)}{1}+1 = 6$$

filter $=4$ output shape $= 6\times 6\times 4$

2. $$\frac{(10+2\cdot 1-3)}{1}+1 = 10$$

filter $=2$ output shape $= 10\times 10\times 2$

3. weights per filter $= 3\times 3\times 1 + 1$ bias $= 10$

$3\times 10 = 30$ trainable parameters.

4. $X = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 2 & 1 & 0 \end{bmatrix}$ $K = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ $s = 1$

5. $$\begin{bmatrix} \begin{bmatrix} 1 & 3 \\ 4 & 2 \end{bmatrix}\begin{bmatrix} 2 & 0 \\ 5 & 1 \end{bmatrix} \\ \begin{bmatrix} 0 & 2 \\ 1 & 6 \end{bmatrix}\begin{bmatrix} 7 & 3 \\ 4 & 2 \end{bmatrix} \end{bmatrix} \overset{\max}{=} \begin{bmatrix} 4 & 5 \\ 6 & 7 \end{bmatrix}$$

6. Filters are used to detect certain features in the input image such as: edges, curves, etc.

7. Padding ensures the edges of input are properly analyzed also it helps to have the output have same dimensions as input.

8. Max pooling down samples the image and keeps maximum values.